PROJECT 3

Bank Customer Churn Prediction and KPI Dashboard

Project documentation | Data Analytics Internship

1. Project Overview

This project analyses bank customer churn, builds machine-learning models to estimate churn risk, and presents customer and churn insights in Power BI and Tableau Public. The goal is to help a bank identify customer groups that may need retention attention.

2. Dataset

Dataset: Bank Customer Churn (Maven Analytics Data Playground). The original dataset contains 10,000 customer records and 13 columns. The target column is **Exited** (1 = churned, 0 = not churned).

Source: Maven Analytics – Bank Customer Churn
Download date: add the actual date on which the dataset was downloaded. Review the provider's current usage and licensing terms before redistribution.

3. Data Quality and Preparation

- Loaded and inspected 10,000 records and 13 columns.
- Checked missing values and duplicate rows; none were found.
- Checked the target values and basic ranges for credit score and age; no invalid values were identified in these checks.
- Retained CustomerId for joining and reporting, but excluded CustomerId and Surname from model predictors.
- Prepared numeric and categorical features using scaling and one-hot encoding within a preprocessing pipeline.

4. Exploratory Data Analysis

Finding	Observed result
Overall churn	2,037 of 10,000 customers (20.37%) churned.
Geography	Germany had the highest churn rate: 32.44%; France 16.15%; Spain 16.67%.
Gender	Female customers: 25.07%; male customers: 16.46%.
Membership	Inactive members: 26.85%; active members: 14.27%.
Products	Churn was 27.71% for one product, 7.58% for two, 82.71% for three, and 100% for four. The three- and four-product groups are small, so interpret these rates cautiously.

5. Machine-Learning Models

A stratified 80/20 train-test split was used. Logistic Regression and Decision Tree models were trained in preprocessing pipelines. A class-weighted Decision Tree was also tested to improve detection of the minority churn class.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	80.80%	58.91%	18.67%	28.36%	77.48%
Decision Tree	85.60%	78.74%	40.05%	53.09%	84.23%
Balanced Decision Tree	77.05%	46.15%	76.66%	57.62%	84.03%

The balanced Decision Tree was used for full-dataset scoring because its higher recall can identify more potential churners, although it also produces more false positives. A probability threshold of 0.60 was selected as a practical operating point for the dashboard. Threshold tuning was explored using the holdout results, so these threshold-specific results are exploratory and may be optimistic; a separate validation set would be preferable for formal threshold selection.

6. Dashboard Deliverables

- Power BI: KPI cards for total customers, actual churned customers, actual churn rate, and predicted high-risk customers; charts for churn by geography, gender, membership status and number of products; risk distribution; and high-risk customers by geography.
- Power BI what-if scenario: offer acceptance rate is adjustable. Estimated retained customers use an assumed 40% success rate among offer accepters. This is a scenario estimate, not a measured campaign result.
- Tableau Public: overall churn rate, churn by geography, gender and active membership, and customer churn-risk distribution.

7. Key Recommendations

- Review customer experience and retention opportunities in Germany, where the observed churn rate was highest.
- Investigate reasons for higher churn among inactive members and consider suitable re-engagement campaigns.
- Examine differences in churn by gender and product count without assuming these attributes cause churn.
- Use the model's high-risk list as a prioritisation aid, not as an automatic decision. Validate customer circumstances before contacting or making offers.

8. Limitations

- The dataset is a snapshot and may not represent current customer behaviour.
- Associations in exploratory analysis do not prove causation.
- The class-weighted model trades some precision and accuracy for higher recall.
- The retention-offer calculation depends on an assumed 40% success rate and is not evidence of actual campaign impact.
- Threshold selection should be validated on separate data before operational use.

9. Repository Deliverables

- customer_churn_analysis.ipynb – analysis, preprocessing, model training and evaluation.
- cleaned_customer_churn.csv and churn_predictions.csv – prepared data and scored customer records.
- model_evaluation.md and data_cleaning_log.md – supporting reports.
- Customer_Churn_Dashboard.pbix – Power BI dashboard.
- Tableau workbook, published dashboard URL and dashboard screenshots.

10. AI Tools and Verification

ChatGPT was used for learning support, code explanations, troubleshooting and documentation drafting. The project owner should review this statement and edit it if it does not accurately describe their use. Code was run in the notebook and outputs were checked against the dataset and reported metrics. All final calculations, charts, assumptions and documentation should be independently reviewed before submission.

Tableau Public link: paste the published dashboard URL here before final submission.